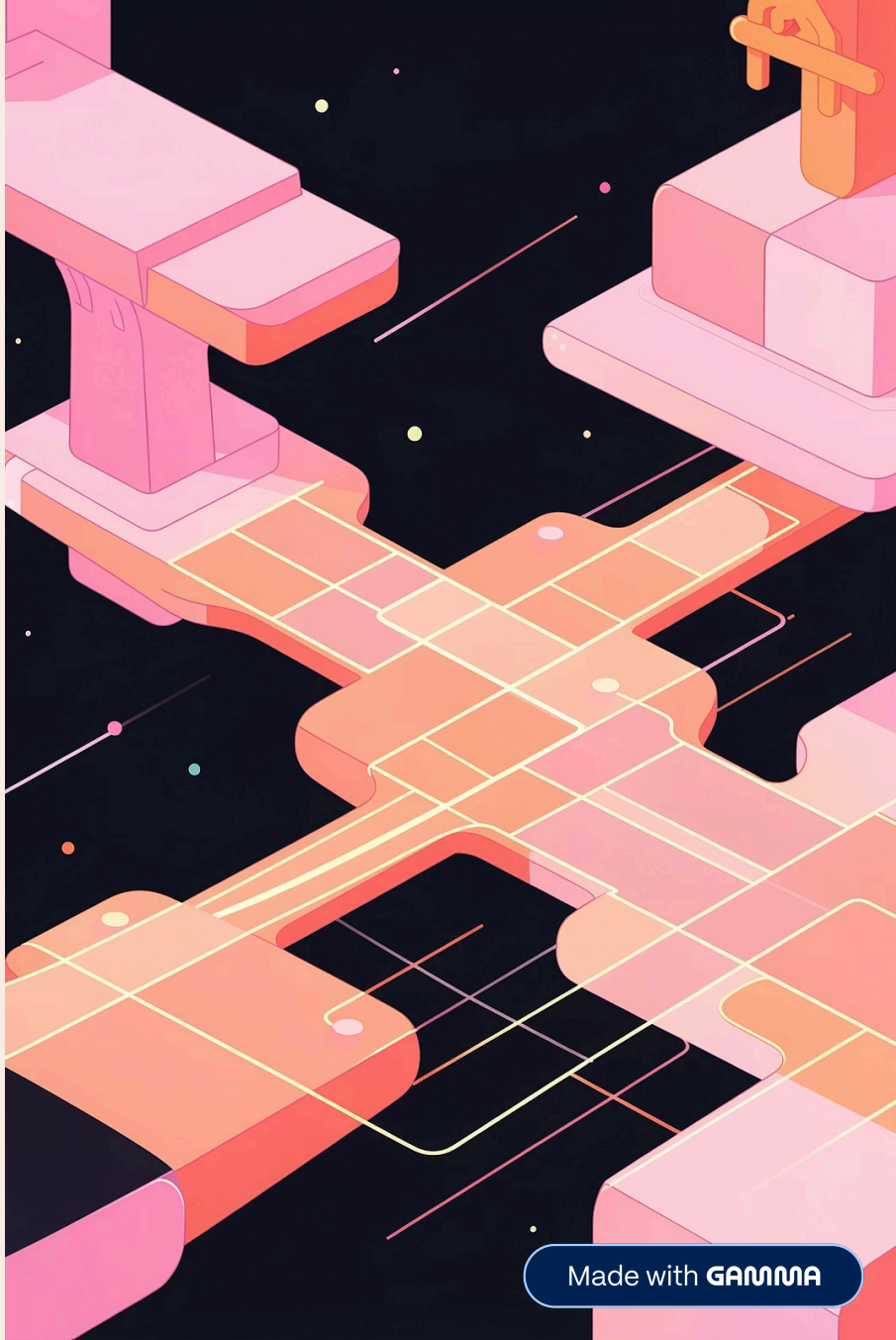
GLDtk

Generative Level Design Toolkit

From a sentence to a playable level.



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Level Design Is Broken

⏳ Slow

A single polished level takes days of iteration. Studios ship fewer levels than players expect.

💸 Expensive

Senior level designers are rare and costly. Most teams can't scale their creative output.

🔁 Repetitive

Without fast iteration, studios fall back on the same layouts. Players notice.

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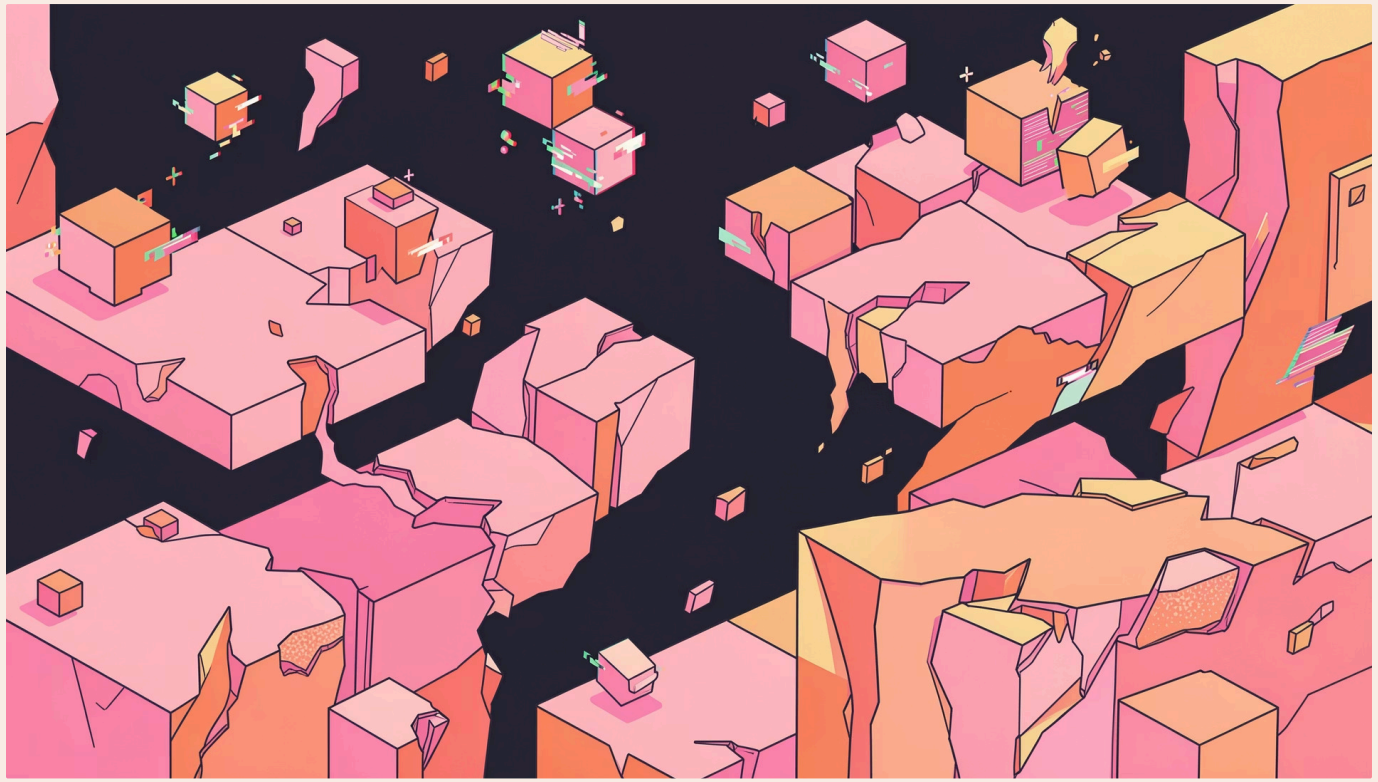

THE AI TRAP

AI Alone Isn't Enough

Existing AI tools generate levels that **look** creative — but are completely unplayable.

Hallucinated Geometry

Jumps that can't be made. Exits that can't be reached. Pure neural AI has no concept of physics.

Zero Playability Guarantee

Without a physics model, every output requires manual QA — defeating the speed benefit entirely.

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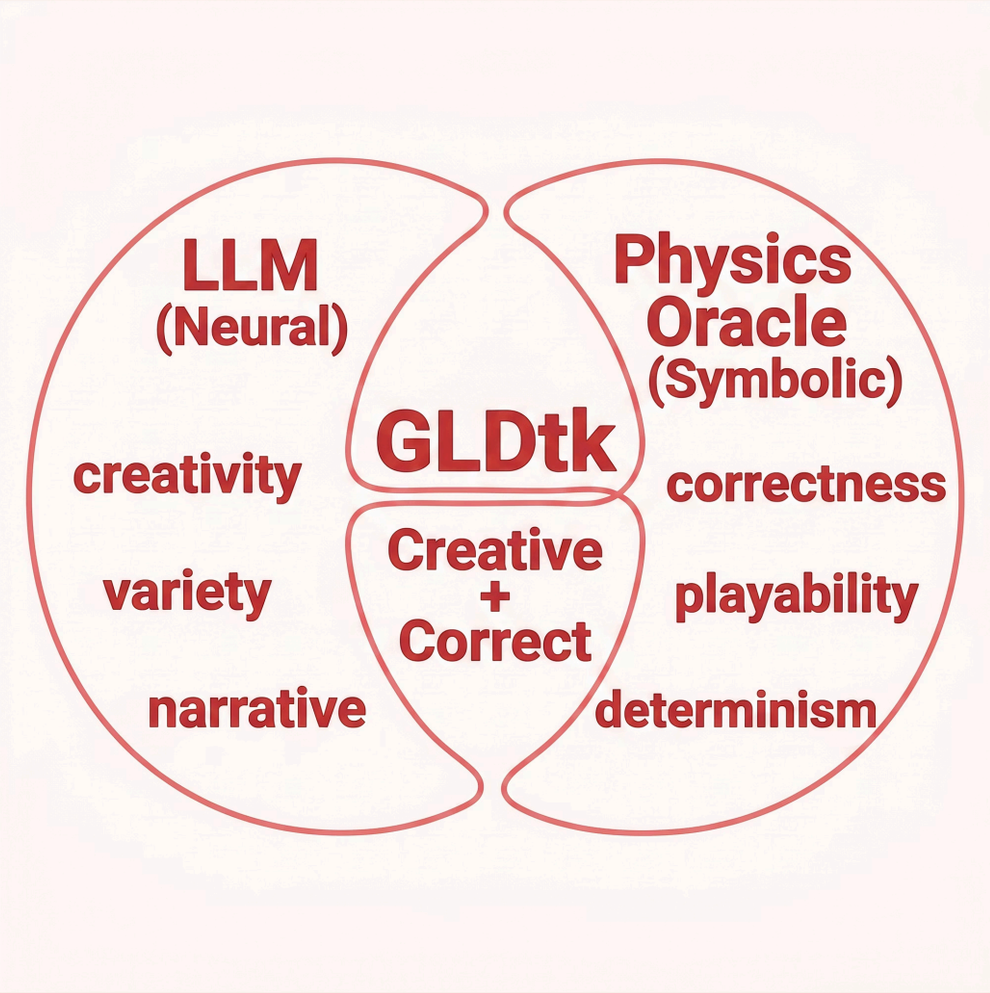
Our Solution: Neuro-Symbolic AI

GLDtk combines the creative power of large language models with the rigor of symbolic physics — so you never have to choose between variety and playability.



LLM — Neural

Creativity, variety, and narrative depth. Generates rich level graphs from natural language.

Physics Oracle — Symbolic

Correctness, playability, and determinism. Validates every jump with kinematic equations.

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PIPELINE

How It Works

From idea to playable level in seconds — with physics validation at every step. No manual QA required.

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The Physics Oracle

Every jump is validated with a single kinematic equation — **instantly and deterministically**.

The Equation

$D = v^2 - 2g \cdot \Delta y$ — if $D < 0$, the jump is physically impossible and rejected instantly.

Millisecond Fast

100% deterministic. No neural inference overhead. Zero human labeling needed.

Always Correct

The oracle is the ground truth. Every level that passes is guaranteed playable.

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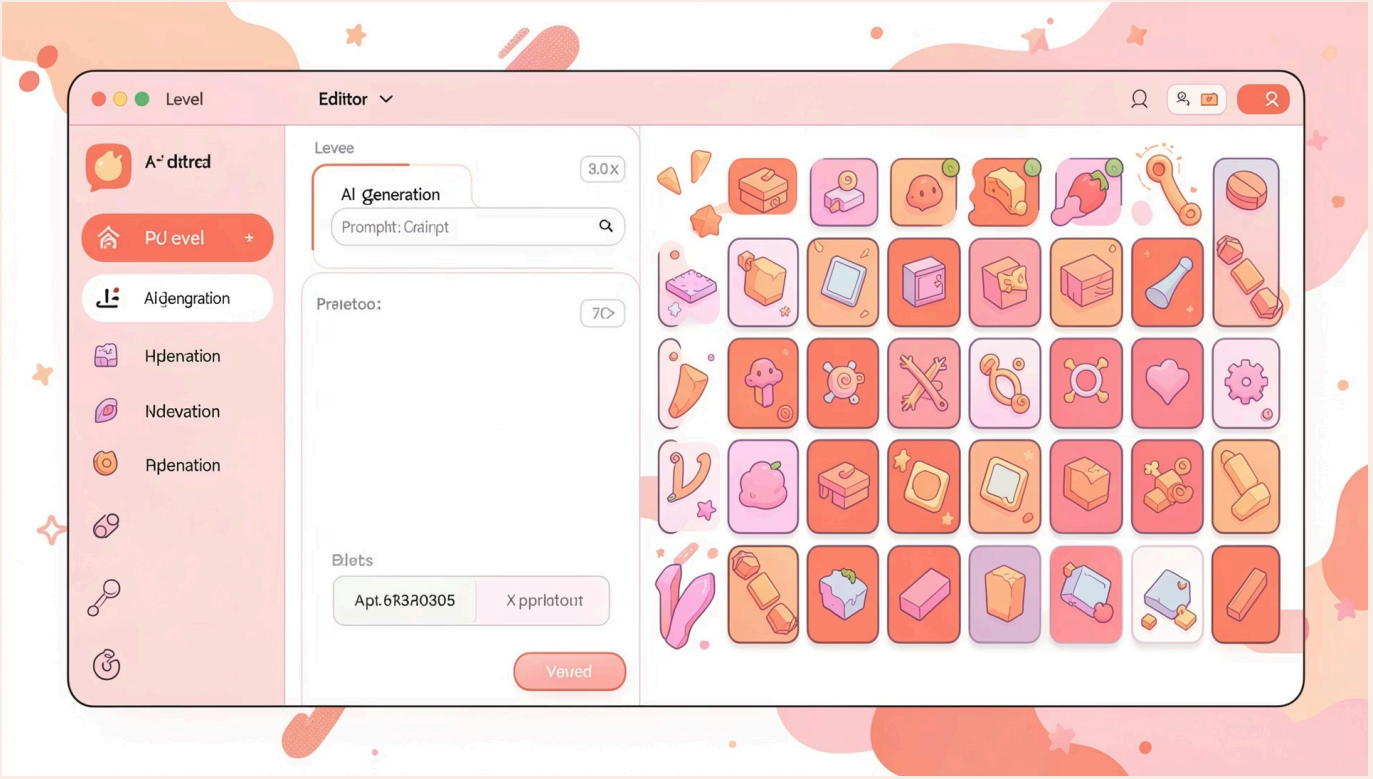

INTEGRATION

Built Into the Editor

GLDtk lives inside **LDtk** — the industry-standard 2D level editor. No context switching. No file management.

Type a prompt

Describe the level in plain English — "a vertical tower with moving platforms and a hidden exit."

Click Generate

The level appears in your project instantly — validated, playable, and ready to iterate on.

Feels native

Designers don't leave their workflow. It feels like a built-in feature, not a plugin.

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How GLDtk Compares

Pure AI Tools

Speed: Fast

Playability: Not guaranteed

Easy to Maintain: No

Manual Design

Speed: Very slow

Playability: Playable

Easy to Maintain: Difficult

GLDtk

Speed: Fast

Playability: Always playable

Easy to Maintain: Yes

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It Learns Over Time

Every generated level is a **free training example**. The physics oracle acts as the reward signal — no human labels ever needed.

Collect Verified Levels

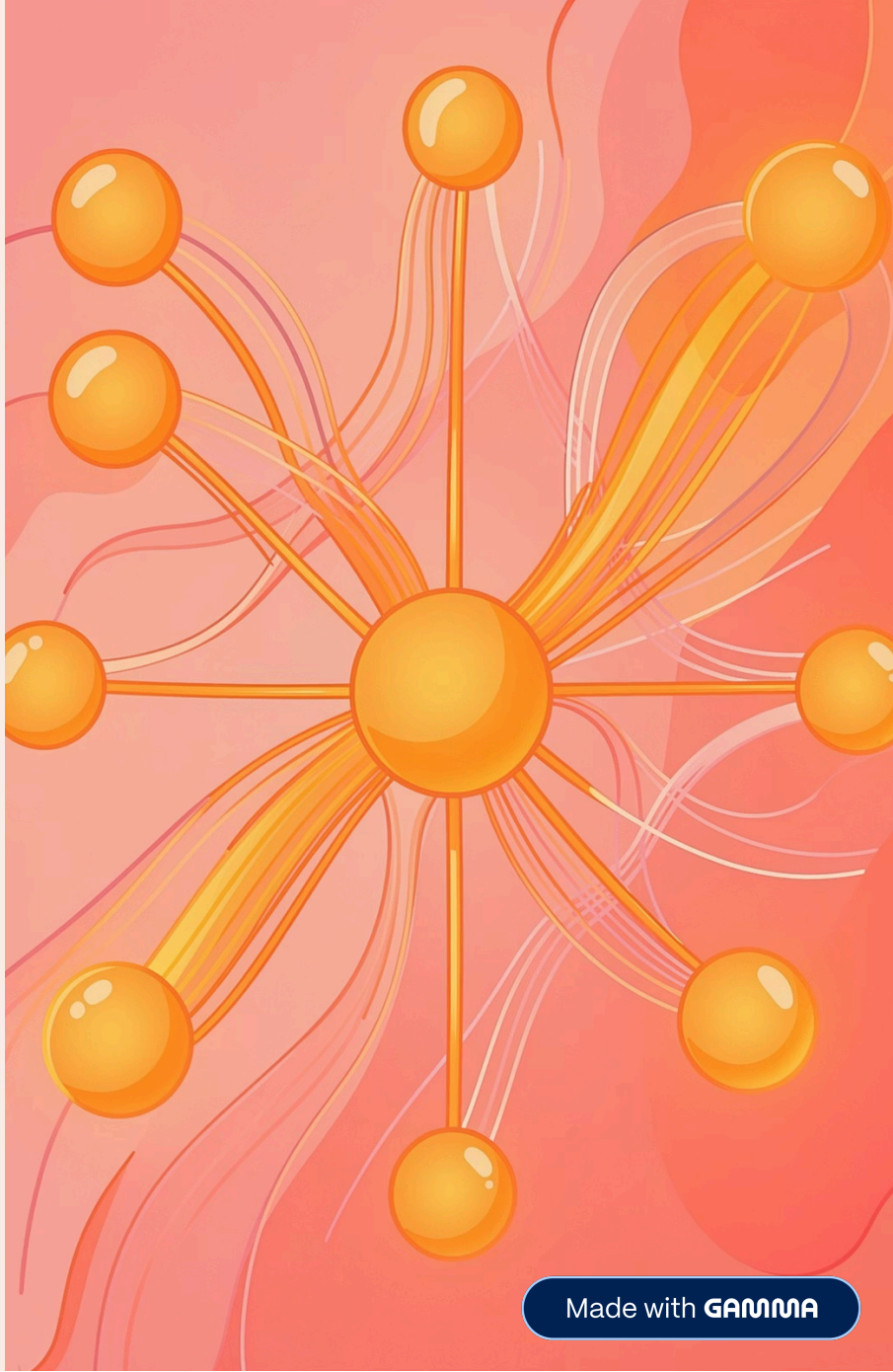
Every level that passes the oracle is a high-quality training sample — automatically labeled.

Fine-Tune Locally

Fine-tune a local model on your studio's verified levels. The model learns your design language.

Offline, Instant, Perfect

Goal: a fully local model generating physics-valid levels with zero API calls and zero latency.



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The Physics Don't Lie.

The Problem

Level design is slow and expensive. Pure AI generates unplayable geometry.

The Solution

LLM for creative ideas + physics oracle for guaranteed correctness — working together in one pipeline.

The Result

Creative, playable levels from one sentence — inside the designer's own tool.

> **GLDtk** — From a sentence to a playable level.

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